A Reader's Map to the Event-Density Gravity Line

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Five documents (openly archived on Zenodo — each link opens that record):

Shorthand	Title	Archived
GR-I *(General Relativity I)*	The Bandwidth Lapse and the Factor of Two	zenodo.org/records/20665981
GR-II *(General Relativity II)*	The Arrow's Fingerprint	zenodo.org/records/20666023
KM-I *(Khronon–MOND I)*	The Arrow's Deep Field	zenodo.org/records/20672818
KM-II *(Khronon–MOND II)*	The Arrow's Horizon	zenodo.org/records/20672870
One Field *(the capstone letter; 1–4 summarized)*	Gravity, Dark Matter, Dark Energy, and the Arrow of Time in Event Density	zenodo.org/records/20678990

Together they are the mechanism by which Event Density (ED) — a discrete, relational substrate with no fundamental spacetime — climbs to a theory that looks and behaves like General Relativity.

But the real story is richer than "an emergent-gravity model." ED does not approximate GR; it lands exactly on the Einstein weak field, then shows precisely where and why it deviates — and the deviation is the arrow of time's fingerprint.

The gravitational class it lands in is called khronometric gravity (the infrared limit of Hořava gravity) — a theory physicists already study.

What ED adds is not standard: the khronon is not postulated but counted, it is precisely the arrow of time made dynamical, and the same single field goes on to carry the dark-matter phenomenology and the dark-energy scale. **One field, one scale, five roles.**

Below is the structural map of how the five documents assemble the bridge:

> **ED $\rightarrow$ Einstein weak field $\rightarrow$ khronometric gravity $\rightarrow$ MOND $\rightarrow$ cosmology $\rightarrow$ Λ.**

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1. GR-I: The Metric Emerges (and it's Einstein's)

Takeaway: ED's bandwidth field b is the spatial metric, and the commitment rate fixes the lapse.

Result:

- $g_{ij} \sim b^{-1}$
- $N^2 \sim b$
- $\Rightarrow g_{00} g_{rr} = -1$ (the Schwarzschild relation)
- $\Rightarrow$ Einstein's weak-field metric, including the factor-of-two light bending.

This is the reason ED can become a metric theory at all. The arrow is already present here — the lapse is fixed by the commitment rate (primitive P11, irreversibility) — but it does not yet show up as a deviation from Einstein, because khronometric gravity coincides with GR in the static weak field. The fingerprint surfaces in GR-II.

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2. GR-II: The Arrow Forces the Gravitational Class

Takeaway: ED is not pure GR. The arrow lives in the law, so the gauge group is reduced.

Result:

- Gauge: foliation-preserving diffeomorphisms
- Modes: 2 tensor + 1 scalar
- Class: khronometric gravity (Hořava IR limit)

Structural passes:

- $c_t = c$ (GW170817 satisfied as an identity)
- universal Lorentz violation (safe from 10^{-20} bounds)

This is where ED becomes Einstein + one scalar — the scalar is the arrow made dynamical.

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3. KM-I: The Deep Field = MOND

Takeaway: The khronon's deep-infrared regime is the MOND sector.

Result:

- $a_0 = cH_0/2\pi$ inherited from ED cosmology
- Deep-field law: $a = \sqrt{}(a_N a_0)$ (matched to the already-derived Combination Rule — forced-given-Paper_030, an embedding, not a derivation from primitives)
- Lensing passes with no vector field
- Solar system screened
- No ghosts (W-sector adds no time derivatives)

This is how ED unifies GR's weak field with MOND's deep field using one scalar field.

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4. KM-II: The Cosmological Sector and the Dark Fluid

Takeaway: The same scalar has a cosmological face — a dark fluid whose sound speed is the free dial.

Result:

- Regulator family $\mathcal{W}(A^2, \Theta)$
- Orthogonality theorem: cosmology is sequestered from statics
- Dust-like clustering requires $c_s \approx 0$
- The near-vacuum soft spot = the CMB/cluster dial
- Late-time constant automatically $\sim H_0^2 c^2/G$

This is the moment ED's scalar is positioned to serve as a dark-matter analogue on cosmological scales — addressable, not discharged: no dial member is chosen and no CMB spectrum is computed.

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5. One Field, One Scale, One Λ

(the capstone letter; the Λ result itself lives in KM-II §7.1)

Takeaway: The khronon's vacuum term and the substrate V1 integral are the same Λ at two levels.

Result:

- Both faces scale as $H_0^2 c^2/G$
- Sign check passes
- EFT constant pinned: $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda \approx -162$
- Substrate integral must hit that number → falsifiable

This is when ED's cosmology closes: Λ is not two things; it is one boundary seen two ways.

(The $\mathcal{W}_0$ derivation is carried in KM-II, and the letter restates it.)

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6. The Big Picture: How ED Reaches GR

ED starts with a discrete substrate whose only special structure is an irreversible arrow of time.

That arrow forces a preferred foliation, which forces a khronon, which forces khronometric gravity, which coincides with Einstein in the weak field, and whose deep field and cosmological field reproduce MOND and Λ.

Or even shorter:

> **The arrow → the khronon → Einstein + MOND + Λ.**

This is why the capstone letter says:

> **One field. One scale. Five roles. Three ways to kill it. One number to miss.**

The One-Field paper is the natural place to start in reviewing the ED–GR connection. The whole claim is there in five pages.